TERTIARY INFOTECH ACADEMY

LAB 05
Control Infrastructure Change and Configuration Drift

AWS Certified SysOps Administrator Associate Training
TGS-2024051413 · Version v6.0

Scenario
A manual console change has made Northstar Retail's production-like environment differ from its approved template.

Objective
Apply K3, K4, A4 and A5 by validating a CloudFormation template, reviewing a change set, detecting drift and using a Systems Manager Automation runbook with verification.

Deliverable
A validated CloudFormation stack, reviewed change set, drift report, reconciliation decision, Automation verification and cleanup record.

Services and tools
CloudFormation · Change Sets · Drift Detection · Systems Manager Automation · AWS CLI

Safety and cost boundary
- Use only the trainer-authorised learner account and Region.
- Replace placeholders and tag learner-owned resources.
- Capture evidence before cleanup.
- Verify deletion and report retained chargeable resources.

Detailed procedure

Step 1 — Validate `operations-stack.yaml` locally and with CloudFormation; inspect parameters, tags, deletion policy and output.
```
aws cloudformation validate-template --template-body file://operations-stack.yaml | tee evidence/template-validation.json
```

Evidence note: _____

Step 2 — Deploy the baseline stack and record its stack ID and output.
```
aws cloudformation deploy --stack-name SysOps-Lab05-REPLACE_SUFFIX --template-file operations-stack.yaml --parameter-overrides Environment=lab
```

Evidence note: _____

Step 3 — Edit only the working copy as instructed, then create—not execute—the change set.
```
aws cloudformation create-change-set --stack-name SysOps-Lab05-REPLACE_SUFFIX --change-set-name Lab05-ReviewedChange --template-body file://operations-stack.yaml --change-set-type UPDATE --parameters ParameterKey=Environment,ParameterValue=lab | tee evidence/change-set-create.json
```

Evidence note: _____

Step 4 — Describe the change set. Record every Add, Modify, Remove or Replacement action in `CHANGE-DECISION.md`.
```
aws cloudformation describe-change-set --stack-name SysOps-Lab05-REPLACE_SUFFIX --change-set-name Lab05-ReviewedChange | tee evidence/change-set-review.json
```

Evidence note: _____

Step 5 — Execute the trainer-approved change set and wait for UPDATE_COMPLETE.

```
aws cloudformation execute-change-set --stack-name SysOps-Lab05-REPLACE_SUFFIX --change-set-name Lab05-ReviewedChange
```

Evidence note: _____

Step 6 — Create one safe manual tag drift on the learner-owned resource; record actor and UTC time.

```
aws s3api put-bucket-tagging --bucket REPLACE_BUCKET --tagging 'TagSet=[{Key=Course,Value=TGS-2024051413},{Key=DriftDemo,Value=manual}]'
```

Evidence note: _____

Step 7 — Start stack drift detection, wait for completion and export the resource drift detail.

```
aws cloudformation detect-stack-drift --stack-name SysOps-Lab05-REPLACE_SUFFIX | tee evidence/drift-detection.json
aws cloudformation describe-stack-resource-drifts --stack-name SysOps-Lab05-REPLACE_SUFFIX | tee evidence/drift-details.json
```

Evidence note: _____

Step 8 — Decide whether desired or actual state is correct. Reconcile through the template rather than hiding the drift.

```
open CHANGE-DECISION.md
```

Evidence note: _____

Step 9 — Create and execute `automation-runbook.yaml` against the learner-owned bucket; save the execution output.

```
aws ssm create-document --name SysOps-Lab05-VerifyBucket --document-type Automation --document-format YAML --content file://automation-runbook.yaml
aws ssm start-automation-execution --document-name SysOps-Lab05-VerifyBucket --parameters BucketName=REPLACE_BUCKET | tee evidence/automation.json
```

Evidence note: _____

Step 10 — Capture change-set and drift evidence as JPEG screenshots, run validation, then clean up.

```
bash validate.sh
bash cleanup.sh | tee evidence/cleanup.txt
```

Evidence note: _____

Verification

The change set is reviewed before execution, drift evidence identifies the changed property, the decision record explains reconciliation, the Automation runbook verifies state, and validation emits LAB 05 PASS.

Acceptance criteria

- No direct unreviewed production-style change
- Replacement actions explicitly reviewed

- Manual drift is confined to learner-owned resource
- Reconciliation decision cites evidence
- Automation has bounded permissions and explicit output

Evidence checklist

□ Template validation

□ Change-set review

□ Drift details

□ Automation execution

□ completed CHANGE-DECISION.md

□ JPEG evidence

□ cleanup.txt

Troubleshooting

- Check identity, Region and resource ID before changing permissions.
- Resolve every REPLACE placeholder.
- Preserve error output as evidence.
- Escalate cleanup failures immediately.